Yang Du | CS

University of Michigan – Ann Arbor – MI, United States

+1 (734)747-3879 • duyung@umich.edu • young-du.github.io

Final year undergraduate at U-M pursuing a major in Computer Science and minor in Mathematics, applying for Ph.D. in Computer Science. Research interests: theoretical computer science, cryptography, blockchain.

Education

University of Michigan (Ann Arbor) — Michigan, United States
B.S.E. in Computer Science — Major GPA: 4.0/4.0 — *Sep, 2019 - May, 2021*

Shanghai Jiao Tong University — Shanghai, China
B.S.E. in Electrical and Computer Engineering — Major GPA: 3.88/4.0 — *Sep, 2017 - Aug, 2019*

Research Projects

Oblivious Data Structure
Guide: Prof. Elaine Shi, CS Dept, Carnegie Mellon University — *Apr, 2020 - Present*

- Implement an oblivious AVL tree data structure with Path ORAM and deterministic eviction.
- Find and compare efficiency of oblivious range query data structure.

Efficient Algorithm to find number of prime factors of Integer with Oracle
Guide: Dr. Ilya Volkovich, CSE Dept, University of Michigan — *April, 2020 - Sep, 2020*

- Devise the first efficient deterministic algorithm for approximating $\omega(N)$ the number of prime factors of an integer given in addition oracle access to Euler's Totient function.
- Show that the algorithm can be extended to handle a more general class of additive functions that "depend solely on the exponents in the prime factorization of an integer"
- Gives the first algorithm that approximates $\omega(N)$ without necessarily factoring N.

Secure Cloud Manufacturing
Guide: Prof. Z. Morely Mao, EECS Dept, University of Michigan — *Sep, 2020 - Present*

- Cybersecurity implementation, cloud system design, database creation and management, data analytics and verification

Free Speech Technology
Guide: Prof. Dick Robert, EECS Dept, University of Michigan — *Dec, 2019 - May, 2020*

- Designed to support free speech by preventing blocking, censorship, and surveillance.
- Support communication during events that disable the internet and cellular communication infrastructures.

Publication

Yang Du, Ilya Volkovich, ***Approximating the Number of Prime Factors Given an Oracle to Euler's Totient Function***, ITCS 2021, submitted.

Key Academic Projects

Cryptocurrency and Blockchain
Guide: Prof. Mahdi Cheraghchi, CSE Dept, University of Michigan — *Mar, 2020 - Apr, 2020*

- Study Bitcoin (public key encryption) and Z-cash (zero-knowledge proof).
- Blockchain application to e-voting and supply chain tracking.

Deep Learning Techniques for Classification and Feature Learning
Guide: Dr. Kutty Sindhu Krishnan, CSE Dept, University of Michigan — *Feb, 2020 - Apr, 2020*

- Use Pytorch to classify pictures different types of food.
- Compare different methods: NN, CNN, Autoencoder.

Teaching Experience and Internship

Linear Algebra, Proof Tutor
Instructor: Dr. Scott Schneider, Maths Dept, University of Michigan *Jan, 2020 - May, 2020*
- Hold office hours to discuss homework questions with students.

Foundation of Computer Science, Grader
Instructor: Dr. Amir Kamil, CSE Dept, University of Michigan *Sep, 2019 - May, 2020*
- Grade homework and exam paper.

Honors Mathematics II, Teaching Assistant
Instructor: Dr. Horst Hohberger, Shanghai Jiao Tong University *Sep, 2019 - May, 2020*
- Lead weekly discussion session.
- Grade homework and exam paper.

Software Engineer of Developing a Teaching Management System
Shanghai Suxun Education Ltd. *Dec, 2018 - Feb, 2019*
- Front end(Android) development.
- Design and format API and design data structure to minimize data transferred.

Scholastic Achievements

- **Roger King Scholarship** awarded by University of Michigan *Aug, 2020*
- **Dean's Honors List** awarded by University of Michigan *Dec, 2019*
- **Fuda Scholarship** awarded by Shanghai Jiao Tong University *Nov, 2018*
- **Undergraduate Merit Scholarship** awarded by Shanghai Jiao Tong University *Nov, 2018 & 2019*

Technical Strengths

- **Programming Languages:** C/C++, Python, Java
- **Development:** Android
- **Others:** LaTeX, Matlab, Mathematica, Origin